



Organizational, professional, and patient characteristics associated with artificial intelligence adoption in healthcare: A systematic review

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ABSTRACT

Objectives: This review study was aimed to identify and document organizational, professional, and patient characteristics influencing the adoption of Artificial Intelligence (AI) in healthcare.

Methods: We searched electronic databases for relevant publications and outlined the eligibility criteria based on the study Participants (healthcare professionals), Concept (AI adoption), and Context (healthcare settings). We included peer-reviewed quantitative, qualitative, and mixed-method studies published in English and implemented a qualitative data synthesis approach to categorize and interpret the results.

Results: A total of 27 studies were included in the final analysis. We organized professional characteristics associated with AI adoption in healthcare settings into three main categories: psychosocial, experiential, and background elements. We organized the organizational characteristics into structural and cultural elements. We grouped patient characteristics into three categories: psychosocial factors, background characteristics, and medical characteristics.

Conclusions: Psychosocial factors such as perceived ease of use or usefulness, performance or effort expectancy, and social influence were major influential factors among healthcare professionals and patients. However, the perceived threat to autonomy was negatively associated with AI adoption. Structural factors such as the organization's size, workflow, training, and security play a crucial role in adopting AI technologies. A limited number of studies cited patient's demographic (e.g., age, sex, and race) and medical (e.g., the severity of illness and historical procedures) characteristics as factors associated with AI adoption. Further studies are needed to better understand the mechanisms of AI adoption in healthcare organizations and settings.

Introduction

Artificial Intelligence (AI) in healthcare is described as developing technology systems that can manage and analyze data, come to decisions, and support users in the healthcare industry by imitating human cognitive functions [1]. In healthcare settings, AI technologies are classified into two main categories: Machine Learning (ML) and Natural Language Processing (NLP) techniques. ML aims to congregate patients' traits or estimate the disease outcomes by analyzing structured medical data such as genetic and imaging data. In comparison, NLP uses unstructured data such as clinical records and medical journals to augment

structured medical data, which can finally be analyzed using ML techniques [1, 2]. AI in healthcare is adopted to be used by organizations, providers, and patients. AI has broad uses in healthcare and is effectively used in research [3, 4], diagnosis and treatments [1, 5], maintaining patient information [6, 7], clinical documentation [8], or even something as simple as scheduling patient appointments [9].

While the use of AI in healthcare has changed drastically with the increased adoption of technology in certain diagnosis and treatment processes, more expanded use of these technologies in the health sector, including funding and payment mechanisms, institutions, professional practices, and patient relations, are expected in the future. The

Abbreviations: ML, Machine Learning; NLP, Natural Language Processing; TRA, Theory of Reasoned Action; TPB, Theory of Planned Behavior; TIB, Theory of Interpersonal Behavior; TAM, Technology Acceptance Model; JBI, Joanna Briggs Institute; CDSS, Clinical Decision Support System.

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importance of using AI in healthcare is so crucial that by the year 2021, an estimated total public and private sector investment in healthcare AI reaches \$6.6 billion [10]. The rise of AI in healthcare over time can positively impact patient outcomes and organizational efficiency [11]. It is estimated that the healthcare industry could see 20% of unmet clinical demand fulfilled by AI. Some of the main challenges in delivering healthcare services are improving patient outcomes and cutting high costs. With the implementation of AI solutions in healthcare, the industry is estimated to save around \$150 billion annually by 2026 [12].

Despite its advantages in providing efficient and timely services [13, 14], the adoption of AI techniques in healthcare settings faces many challenges and barriers [15]. The main challenge of the effective adoption of AI is its acceptance by healthcare professionals and patients. Whereas professionals' resistance to adopting AI is somewhat attributed to the perceived threat to professional autonomy [16] and future job security [15, 16]. Patients' main concern is privacy threats [17]. There also remain some challenges in terms of organizations that use or are supposed to use AI technologies. While technology can help an organization become more cost-efficient by reducing errors, AI's adoption is an expensive endeavor in itself. If a clinical IT system is implemented and goes unused by physicians who do not accept it, there is a risk of a health system suffering financial repercussions [18]. If healthcare professionals do not learn how to adopt AI, the financial losses to organizations can be devastating as they must adapt to the growing and changing healthcare industry. With the sheer amount of data and discoveries that make up the medical field, organizations need to utilize technology to manage and store the information successfully. The development of AI has transformed the healthcare field, and providers that are not adapting to the change will see the negative financial and quality impacts that will follow [1].

A handful of theories and models explain the acceptance and adoption of technologies. For example, according to the Theory of Reasoned Action (TRA), three main cognitive components including "attitudes (unfavourableness or favourableness of person's feeling for a behavior), social norms (social influence), and intentions (individual's decision to do or don't do a behavior)" can determine any human behavior. Three theories have also emerged on the basis of TRA, including Theory of Planned Behavior (TPB), Theory of Interpersonal Behavior (TIB), and Technology Acceptance Model (TAM). In the TPB model, perceived behavioral control is added to expand the TRA model. Primarily, the availability of resources, opportunities, and skills, as well as the perceived significance of these elements, determine the perceived behavioral control. While both TPB and TRA models assume that an individual's behavioral intention influences their behavior, TPB considers the perceived behavioral control for an individual's actions. TIB model, on the other hand, highlights the effect of social and emotional factors on the human's behavior complexity. Hence, TIB not only includes all characteristics of TRA and TPB but also improves the behavior prediction power by adding factors such as habits and facilitating conditions. TAM assumes that three factors, including perceived usefulness, perceived ease of use, and attitude toward the use of technology, determine users' motivation. Sometimes, factors such as "user training, system characteristics, user participation in design, and the implementation process nature" are considered in the TAM model as external variables. There are also other theories in terms of technology acceptance and adoption; however, TAM is known as one of the most cited models [19].

The primary rationale for implementing technology in a clinical setting is improving patient solutions and cutting costs within an organization. Some AI characteristics such as ease of use, early diagnosis and treatment, efficiency, and reducing medical and administrative errors are appealing [20, 21]. With improved diagnostic methods and clinical documentation through AI, we can see improved patient outcomes and medical records' thoroughness. Providers and healthcare consumers that are hesitant to adopt AI can be identified across a range of demographics. Some previous studies identified the individuals' age and

sex, and organizational capabilities as potential AI adoption predictors [22, 23]. Many physicians who oppose the technology tend to believe that it is a threat to their professional autonomy. Professional autonomy is broadly defined as professionals having control over their own work according to their individual judgment, knowledge, and expertise [18]. In line with barriers to AI adoption in healthcare, studies have cited practice standardization and user training to facilitate AI acceptance and adoption [24].

Although there are a handful of studies explored the barriers and facilitators of the adoption of AI in healthcare [25–28] and its advancement [29], it is not clear what individuals (healthcare professionals and patients) and organizations factors attribute to AI adoption or acceptance in healthcare settings. Though meaningful implementation of ML and other AI techniques in health care depends to some degree on support from patients and healthcare professionals [25], little is known about the influential factors related to adopting these techniques. For instance, an integrative review aimed to explore the healthcare professionals' understanding and experiences of using AI in healthcare provision found only one study eligible to review [30]. This review study aimed to systematically identify and document the professional, organizational, and patient characteristics associated with AI adoption in healthcare settings.

Methods

We used Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) guidance in designing the study and outlining the final report [31].

Adoption of AI technologies in healthcare not only could be explored by using different theories, as is mentioned earlier in the introduction, but also can be explained from various perspectives, including condition, technology, value proposition, adopters, and organization. Each of these aspects is divided into smaller elements. The condition is related to the clinical and sociocultural aspects of the illness and comorbidities. The technology is related to the material and technical features of technology of use, which in our study is AI-assisted services. The value proposition is concerned with the values that technology can create. These values could influence both supply and demand sides. The adopter system includes professionals and clients (patients and caregivers). The organization aspect is related to the organization's capacity and readiness to initiate or adopt new technologies and innovations [32]. A framework that shows an overview of the factors that can influence AI technologies in a broad concept is adopted from the study of Greenhalgh et al. [32] and is presented in Fig. 1. Each of these aspects is wide in nature and can influence the production, marketing, acceptance, and consistent adoption of AI technologies in a broad context of healthcare. In order to have a narrower view on factors that are attributed to the adoption of AI technologies at a level in which healthcare services are provided for patients, such as hospitals and other medical facilities, we merged some aspects and focused on three main domains including organizational, professional, and patient factors. Fig. 1. Shows contributing factors to the adoption of artificial intelligence in the healthcare sector. The main research questions include:

- What are the organizational characteristics associated with the adoption of AI in healthcare?
- What are the professionals' characteristics associated with the adoption of AI in healthcare?
- What are the patients' characteristics associated with the adoption of AI in healthcare?

Eligibility criteria

We developed Population, Concept, and Context (PCC) and types of the eligible studies using the Joanna Briggs Institute (JBI) Manual for

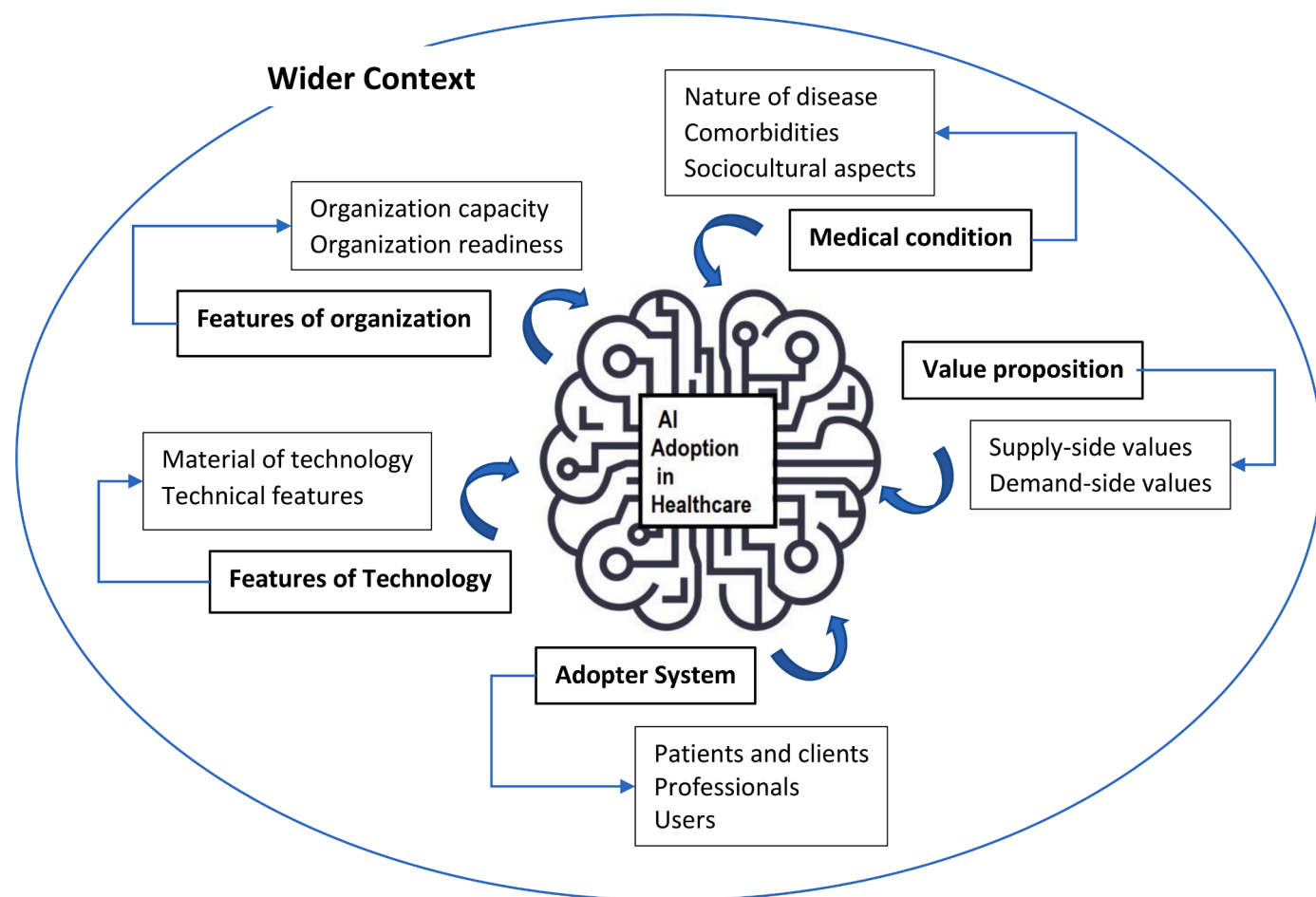


Fig. 1. Contributing factors to the adoption of artificial intelligence in the healthcare sector.

Evidence Synthesis [33] explained below.

Population or Participants: Healthcare professionals who provide preventive, diagnostic, treatment, or rehabilitation services for people in need, regardless of their discipline, work experience, age, sex, and healthcare organization where they practice.

Concept: Adoption of AI as the desirable intervention or concept.

Context: Any healthcare settings, including primary or secondary care, private or public, outpatient or inpatient.

Types of studies: English language, peer-reviewed quantitative studies including experimental, observational, and cross-sectional studies, as well as qualitative and mixed-method studies. Review studies were considered for reference checking purposes.

Exclusion criteria

We excluded the studies that have explored the factors associated with IT adoption without referring to AI techniques. Moreover, we excluded the studies on AI medical devices used in settings other than healthcare.

Databases and search strategies

After conducting a pilot search and using Medical Subject Heading (MeSH), we identified key terms in three main categories to outline our search strategies: *users or beneficiaries*, *artificial intelligence*, and *adoption* (see Table 1). We searched independently for each category. Next, we searched combining all three categories to ensure that the search strategy was suitable to address the research questions. Appendix Table 1. includes search terms specific to each database.

Table 1

Search terms and search strategy.

Category 1 Users or Beneficiaries	Category 2 Artificial Intelligence	Category 3 Adoption or resistance
Patient* OR health OR Medical OR Hospital*	"Artificial Intelligence" OR "Machine Learning" OR "Natural Language Processing" OR "Computational Intelligence" OR "Computer Reasoning"	Adoption OR Acceptance OR Implementation OR Resistance OR approval OR Use OR utilization
Search Strategy: 1 AND 2 AND 3		

Methods of screening and selection criteria

We searched PubMed, Scopus, Embase, and Wiley on February 2, 2020, and updated our search on February 6, 2021. Moreover, we reviewed the reference and citation lists of the relevant studies and gray literature to find any additional relevant studies. Five researchers reviewed the remaining studies' titles and abstracts based on eligibility criteria after pulling the search results into Endnote X9 software program and removing duplicate results. Any disagreements among the investigators were discussed and solved by the lead investigator. Two investigators independently reviewed the full texts of the relevant studies in the next step. Before starting the main data extraction practice, the two investigators implemented the pilot extraction of data from three papers to ensure that the items presented in the extraction table were suitable and in line with the research questions. Finally, the two

investigators extracted the information using the data extract sheet designed based on the JBI extraction table format [33] in advance. This datasheet was confirmed to be suitable for this review purpose in the pilot test.

Data synthesis

We classified results into three major categories and sub-categories by identifying the central concept and priori. Then, we presented the main findings in a narrative format based on affinities found between the results extracted from studies included in the review. Moreover, to summarize the findings, we presented the results in a tables and figures.

Results

Fig. 2 shows the flow of the search and review process. Of 3258 results from the database search and other sources, 2292 records were remained after removing the duplicated results. We reviewed these remaining studies based on the title and abstract and identified 37 papers for further full-text review. Twenty-seven studies met eligibility criteria, and ten studies were excluded due to theoretical nature and irrelevant context. We outlined the findings in three main categories associated with AI adoption in healthcare: *professional*, *organizational*, and *patient* characteristics, and summarized those characteristics in

Table 2.

Although studies were from both developing and developed countries, most of the studies were from the United States ($n = 9$). The studies were focused on various types of AI technologies, including clinical decision support system (CDSS), clinical information system (CIS), intelligent health monitoring system (IHMS), electronic medical records (EMR), robotic-assisted medical services, automated pain recognition devices, symptom assessment app, ophthalmic devices, and clinical reminder system. Many of the studies collected data by using 5-point Likert scale questionnaires in cross-sectional surveys. Six qualitative studies and one mixed-method study were also included in the review. We categorized the results in three main headings and eight sub-headings illustrated in Fig. 3. The complete data extraction forms are included in Appendix Table 2.

Professional characteristics

We categorized professional characteristics associated with AI adoption in healthcare settings into three main categories, including psychosocial factors, experiential factors, and background characteristics.

Psychosocial factors

Most studies have explained the psychosocial factors' association

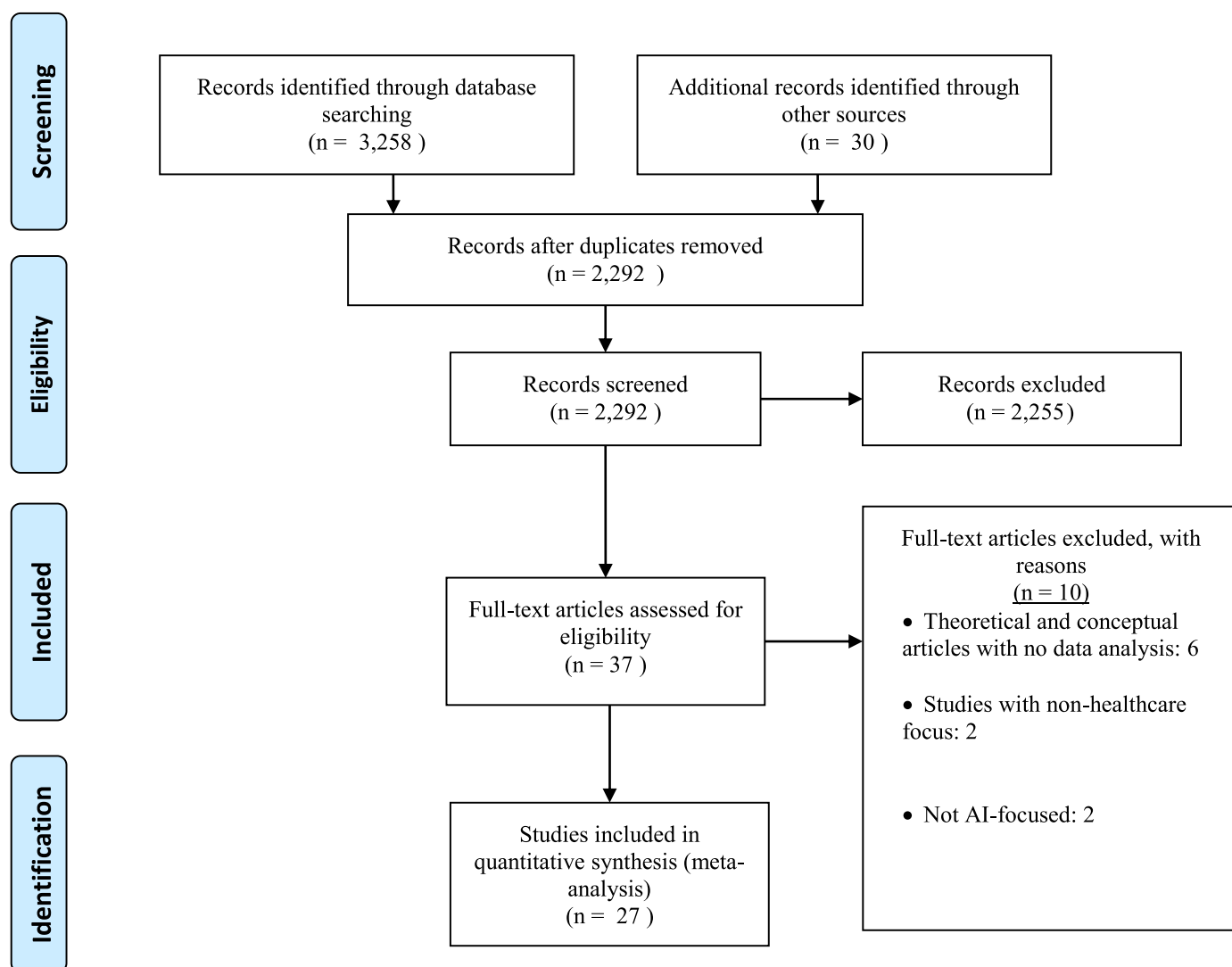


Fig. 2. PRISMA flow diagram for article selection process in this systematic review.

Table 2

Professional, organizational, and patient characteristics contributing to the adoption of artificial intelligence in healthcare.

Inflential factors	Study (First author's last name, publication year)
PROFESSIONAL FACTORS	
Psychosocial factors	
Perceived performance/effect expectancy	Esmailzadeh, 2015; Shibl, 2013; Chang, 2007; Tseng, 2013; Dünnebeil, 2012; Sambasivan, 2012; Chang, 2007; Fan, 2018
Perceived endeavor/effort expectancy	Tseng, 2013; Dünnebeil, 2012
Perceived usefulness	Walter, 2008; Johnson, 2014; Benmessaoud, 2011; Dünnebeil, 2012; Petitgand, 2020
Perceived ease of use	Walter & Lopez, 2008; Johnson, 2014; Dünnebeil, 2012; Benmessaoud, 2011; Petitgand, 2020
Perceived threat to professional autonomy/ Perceived behavioral control	Esmailzadeh, 2015; Walter & Lopez, 2008; Benmessaoud, 2011; Lin, 2012; Sambasivan, 2012
Attitude toward using technology (trust)	Benmessaoud, 2011; Egea, 2011; Strohm, 2020; Shibl, 2013; Fan, 2018
Social influence	Chang, 2007
Experiential factors	
Experience of using IT/computer	Shibl, 2013; Johnson, 2014
AI/Computer knowledge	Johnson, 2014; Dünnebeil, 2012
Frequency of use	Gardner, 1994;
Perceived inequity	Lin, 2012
Extrinsic motivation among nonusers	Benmessaoud, 2011
Usage group membership	Johnson, 2014
Perceived importance of utility	Dünnebeil, 2012; Watson, 2020
Knowledge on clinical feature that bring a result in machine learning	Brown, 2020
Background characteristics	
Age	S. Walter, 2020
ORGANIZATIONAL CHARACTERISTICS	
Structural factors	
Facilitating conditions: workflow, training, security and integration	Shibl, 2013; Chang, 2007; Benmessaoud, 2011; Petitgand, 2020
Size of the organization	Lamberti, 2019; Watson, 2020
Lack of skilled staff	Lamberti, 2019; Watson, 2020
Budget constraints	Lamberti, 2019; Watson, 2020
Incompleteness in health care data quality	Watson, 2020
Cultural factors	
Professionals' level of involvement in decision makings	Esmailzadeh, 2015; Sambasivan, 2012
Institutional trust	Egea & González, 2011
Attitude toward the machine learning model	Watson, 2020
PATIENT CHARACTERISTICS	
Psychosocial factors	
Sense of uniqueness	Longoni, 2019
Social Influence	Tseng, 2013; Ye, 2019
Perceived performance/effect expectancy	Tseng, 2013
Perceived endeavor/effort expectancy	Tseng, 2013
Perceived risk of harm	Nadarzynski, 2019; Ye, 2019
Perceived usefulness	Ye, 2019
Level of certainty about the quality, trustworthiness and accuracy	Nadarzynski, 2019; Ye, 2019
Perceived behavior control	Ye, 2019
Level of health consciousness	Ye, 2019
Background characteristics	
AI knowledge	Yang, 2019; Lennartz, 2021
Gender	Yang, 2019; Xiang, 2020
Age	Miller, 2020
Race	Yang, 2019
Type of job (healthcare none-healthcare workers)	Xiang, 2020
Education	Xiang, 2020

Table 2 (continued)

Inflential factors	Study (First author's last name, publication year)
Medical characteristics	
Historical procedures	Yang, 2019
Type of condition for consultant	Yang, 2019
Having previous experience of diagnostic errors	Meyer, 2020
Severity of illness	Lennartz, 2021

with AI adoption using the TAM, which explains the professionals' voluntary adoption of IT and testing it in healthcare settings. According to TAM, professionals' attitudes and perceptions regarding a technology could attribute to the adoption of that technology [34, 35]. At this regard, two main determinants of professionals' intention to adopt a technology include perceived usefulness and perceived ease of use [18, 24, 34, 36, 37]. The more useful and easier to use the professionals find an AI-aided technology, the more likely they are to use that technology. Two other psychosocial factors, which most of the studies found as strong determinants of professionals' intention to adopt AI, were described by effort or performance expectancy theories [24, 38-43]. Performance expectancy is the extent to which the professionals perceive that adopting technology will improve their job performance. When healthcare professionals believe that AI will help them achieve improvements in their job, they are more likely to adopt AI. Effort expectancy, on the other hand, is associated with how easy the professionals find using an AI-aided technology. Therefore, usefulness and ease of use are tailored to performance or effort expectancy as well, which all are strong predictors of AI adoption by professionals in healthcare settings.

Perceived behavioral control was another psychological factor that five studies referenced as an determinant of adopting AI among professionals, especially physicians [18, 34, 38, 42, 44]. Although a qualitative study reported factors such as perceived ease of use, perceived usefulness, and perceived behavioral control as both barriers and facilitating factors. Most participants in this study cited these factors as the main barriers to using AI in healthcare settings [34]. Several studies found that physicians or other healthcare professionals' perceived threats to their professional autonomy play a critical role in adopting AI [18, 38, 42, 44]. One study found social influence as another psychosocial factor that might positively influence users' intention to use the AI technologies [40]. Professionals' intention to adopt AI depends on the degree to which they believe how important it is that others believe they should adopt AI technologies such as Clinical Decision Support System (CDSS) [40].

Experiential factors

Experiential factors in this review refer to all factors that are derived by experiences in the field of healthcare. Three studies have identified the experience of using IT or computer [36, 39] and AI or computer knowledge [24, 36] as the main factors associated with AI adoption by professionals. Two studies found computer experience to influence ease of use [36, 39]. One study showed that computer knowledge negatively affected the average use of AI technologies such as clinical reminder systems and electronic health record systems and satisfaction with these services [36]. In contrast, another study showed that knowledge about electronic health services improves the perceived ease of use, which may consequently result in intention to adopt those services [24]. Studies also identified other experiential factors associated with the adoption of AI by professionals at some level, including frequency of use [45], trust in the knowledge base [39], perceived inequity [44], extrinsic motivation among nonusers [34], usage group membership [36], perceived importance of documentation, perceived importance of data security, and perceived importance of its utilization [24, 46].

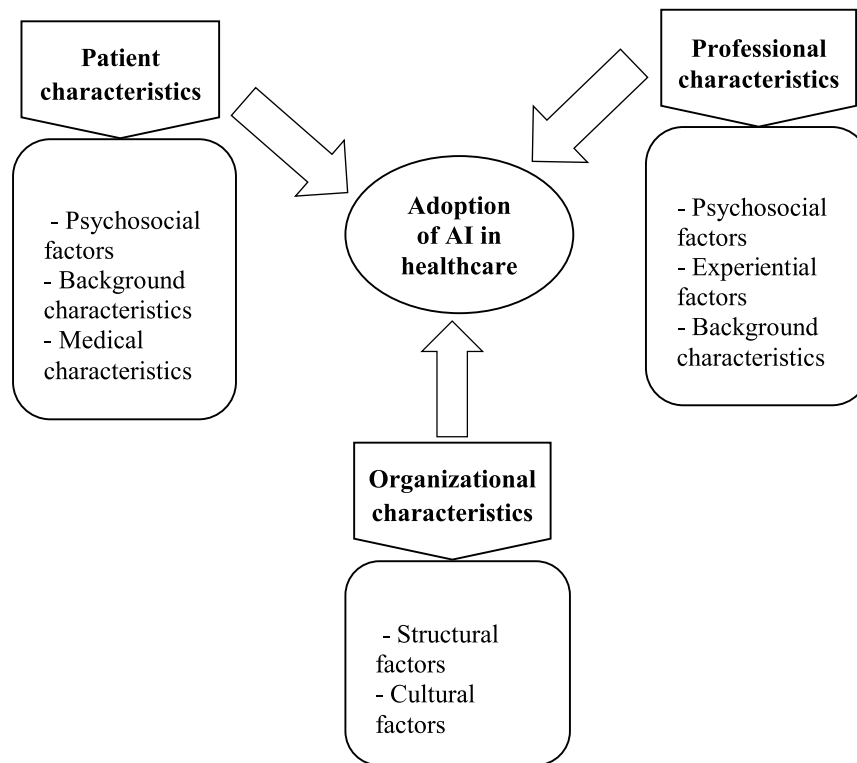


Fig. 3. Organizational, professional, and patient characteristics associated with the adoption of artificial intelligence in healthcare.

Background characteristics

In terms of the background characteristics of professionals that could determine AI-aided technology adoption, we found just one study. This study cited age as a factor contributing to the adoption of AI [45], showing that most of the younger professionals intended to use an automated pain recognition system in healthcare. Included studies did not report other background characteristics of professionals as determinants of the adoption of AI.

Organizational characteristics

We categorized organizational characteristics associated with AI adoption in healthcare settings into two main categories: structural and cultural.

Structural factors

Structural factors include facilitating conditions, integration, organization size, lack of skilled staff, and budget constraints. Among all structural factors, facilitating conditions such as workflow, training, security, and integration [34, 37, 39, 40] appear to play a significant role in adopting AI technologies [34]. Four studies showed that facilitating conditions, including workflow, training, security, and integration, could result in AI adoption in an organization [34, 37, 39, 40]. Additionally, two studies concluded that larger organizations tend to adopt AI more than midsized or small organizations [23, 46].

Cultural factors

We identified three factors, including professionals' level of involvement in decision makings [38, 42], institutional trust [35], and organizational attitude toward the AI techniques [46], as organizations' cultural characteristics associated with the adoption of AI in healthcare settings. Two studies found a positive association between involvement in decision-making and the adoption of AI [38, 42]. One study showed that institutional trust is positively associated with the intention to use electronic healthcare record systems [35]. A qualitative study cited the

attitudes toward the machine learning model as a cultural factor that could hinder this AI technique in healthcare [46].

Patient characteristics

We categorized patient characteristics associated with AI adoption in healthcare settings into three main categories: psychosocial factors, background characteristics, and medical characteristics.

Psychosocial factors

We found a limited number of studies explored the patients' psychosocial factors associated with the adoption of AI describing social influence [47, 48], perceived performance or effort expectancy [41], and perceived risk of harm [48, 49] as primary determinants of patients' intention to adopt AI technologies. A study in the field of ophthalmology showed that the level of eye health consciousness and perceived behavioral control could indirectly have a positive influence on intention to use AI via the mediators' perceived usefulness and perceived ease of use [48]. Another study showed that a sense of uniqueness [47], the degree to which individuals believe that their characteristics are unique and different from others, is a strong determinant of adopting AI technologies in healthcare. This study showed a negative association between clients' sense of uniqueness and AI adoption [47].

Background characteristics

A number of studies cited patients' background characteristics as the main determinants of AI adoption in healthcare [22, 50-52]. One study cited AI medicine knowledge and gender as the main background characteristics of adoption or intention to adopt AI technologies [50]. According to this study, male patients and minorities were more likely to choose AI medical services and were more likely to trust AI medicine than conventional physicians [50]. Another study showed that the receptivity of medical AI was associated with clients' gender and education: males and individuals with a lower level of education reported higher medical AI receptivity [22].

Medical characteristics

We categorized patients' specific medical history, disease severity, and their experience of care as medical characteristics. Three studies highlighted the role of patients' medical characteristics as factors associated with AI adoption in healthcare settings [50, 51, 53]. One study concluded that, in a hospital setting, patients with cancer that had received traditional Chinese medicine, those who had surgery, and those who had never received chemotherapy were more likely to choose AI medical services [50]. Another study highlighted the role of the previous experience of diagnostic errors in adopting AI among the users of a specific AI-assisted symptom checker who were patients with chronic disease. In that study, patients who had previously experienced missed or delayed diagnoses were more likely to use the symptom checker than patients who had not previously experienced diagnostic errors [53]. The third study, which was conducted in a radiology setting and among patients scheduled for computed tomography or magnetic resonance imaging, found the severity of illness as a determinant factor of AI adoption in healthcare settings. According to this study, the acceptance of AI was significantly lower for high severity diseases than diseases of medium severity and low severity [51].

Discussion

We conducted this systematic review to identify the professional, organizational, and patient characteristics that could influence AI adoption in healthcare settings. Due to the articles' lack of congruity, it was challenging to draw a sound conclusion based on our findings. However, there were some commonalities that we identified across the included studies. Overall, there were results that we determined significant on the acceptance of AI, including psychosocial factors, background characteristics of individuals, medical characteristics of patients, and cultural and structural features of organizations.

According to the results, the common determinants of the adoption and acceptance of AI technologies by both professionals and patients were related to psychosocial factors such as perceived ease of use or usefulness, performance or effort expectancy, and social influence. Studies corroborate that these factors could play a significant role in either resistance to or intention to adopt AI technologies by both professionals and clients. For example, while some studies found these psychosocial factors to positively affect the adoption of AI [18, 24, 34], some evidence did not reveal strong proof of the positive effects [35, 43]. Moreover, some other factors, including process orientation and standardization in healthcare practices, can positively contribute to increasing the perceived ease of using AI technologies and, consequently, adopting AI [24]. By considering the theory of TAM, it is not surprising that all these psychological factors could have a negative or positive influence on individuals' intention to adopt AI techniques. This depends entirely on how individuals perceive the results of using these techniques and the extent to which they can trust them.

The results showed that perceived threat to autonomy, or in other words, perceived behavioral control, is one of the main barriers to adoption or acceptance of AI by healthcare professionals and particularly physicians. The use of AI has been shown to improve health outcomes, reduce human errors, and become a reliable tool to help physicians make medical decisions [54, 55]. For instance, before using electronic guidelines, doctors would have to flip through pages of information, waste time, and possibly miss something important that could help a patient. However, not all organizations and physicians have been eager to transition to using AI in their medical facilities due to the threat they may feel to their autonomy. As discussed in two separate studies [18, 38], some physicians are concerned that AI will override or replace their judgment regarding medical diagnoses. Physicians may feel that if they do not follow the system's suggestions and make an error, it results in malpractice.

Moreover, AI systems are imperfect and could potentially mislead a physician's decision. Therefore, physicians need to find a balance

between their judgment and the recommendations made by ML. Every patient is unique and should be treated on an individual need basis. Machine learning and other AI systems use algorithms to determine the common patterns and trends when a patient presents particular symptoms. This is a great tool to standardize treatment, but it should not replace physicians' overall human judgment.

Even though there has been some resistance to the adoption and acceptance of AI by physicians, they will have to adapt to the integration of these systems to keep up with the healthcare industry's demand moving forward. As discussed by Fletcher, hospital competition and the pressure to take on higher volumes of patients can influence the adoption of new technology (like robotic surgery) [56]. Therefore, hospitals that can provide efficient, effective, and cost-saving treatments through AI have a competitive advantage over other medical facilities that are still using traditional methods. Although implementing these new systems can be costly and may take time to incorporate into daily activities smoothly. It has shown to have a good return on their investment and positive long-term outcomes. This could be a reasonable explanation of how organizations' structural characteristics are associated with the successful adoption of AI techniques. As concluded by Lamberti et al., the larger a company is, the more likely it tends to use AI technologies [23], which is not surprising given their larger budgets and enormous structured data.

We found a few studies that explored the public and clients' perception of AI medical services and showed that some patient characteristics could be connected to AI adoption in healthcare settings. The studies showed that patients' background characteristics, including age, gender, and race/ethnicity, are associated with AI adoption in healthcare settings [22, 50, 57-59]. Additionally, although we cannot definitely conclude that in what circumstances and for what procedures patients' medical conditions are associated with the adoption of AI, few studies showed that types of the patients' previous medical procedures and the type of conditions for consultants can influence AI use in healthcare. These examples are of particular value because understanding patients' main characteristics attributed to the adoption of AI could bring insight into how healthcare providers and developers of AI technologies can overcome barriers to the successful use of AI technologies in healthcare. For instance, if race/ethnicity and gender influence the decision to adopt AI in healthcare settings, it is crucial to understand how and to what extent these characteristics could make an effect. Therefore, more studies are needed to contribute to a robust conclusion on what patient characteristics and how they can result in the adoption of AI.

Implications and recommendations

This systematic review is relevant at the current point of time due to the great investments being made to develop and implement AI-aided technologies in healthcare. The findings can help healthcare organizations with a perspective on which users are more resistant to the adoption of AI techniques. From there, decision-makers can use our findings to understand the factors that increase the likelihood of adopting these systems and highlight the importance of AI systems in medical care so that better health outcomes can be made efficiently. This review calls on conducting more researches on organizational and patient factors associated with AI adoption in healthcare settings by considering the medical condition, type of illness, and comorbidities. Additionally, further review studies are needed to identify the political, socioeconomic, and cultural factors as well as the characteristics of the AI technologies that can influence the adoption of AI services in healthcare settings.

Limitations

There are a few limitations that could have restricted the validity and soundness of this study. First, most of the articles included sample

populations constrained to a small geographic area or had an uneven gender ratio. Along with this, there was an overall low-response rate, and most responses were limited to fixed choices as answers. Second, we included only English language studies in the review. Despite these limitations and sample bias, this research on AI use is essential to help healthcare organizations understand the main determinants of AI adoption in healthcare settings. Third, we included organizational, patient, and professional factors that influence the AI adoption in the review and did not study the factors related to the characteristics of the technology or the context in which these techniques are used, including political and socioeconomic environments and cultural settings [32]. Nevertheless, focusing on narrower aspects contributed to the identification of specific factors that can determine the adoption of AI in providing medical services for patients.

Conclusion

The integration of AI in healthcare settings can provide automation of tasks, predictive analytics, and decision-making support. With these improvements, organizations will deliver healthcare faster, with better quality, and at a lower cost. Although challenges still exist, this study was able to identify the main characteristics associated with the adoption of AI by providers, patients, and organizations. Among all factors associated with the adoption of AI in healthcare, most of the studies cited psychosocial factors. There is a large gap in knowledge regarding the patients' characteristics that could influence AI adoption in healthcare settings.

Funding

No funding was received for this study.

Availability of data and materials

The search terms used for pooling relevant published studies and data extract tables are provided as supplementary materials (Appendix Tables 1 and 2).

Code availability

Not applicable.

Ethics approval

This is a secondary review of the published studies. Human subject does not apply to this study.

Consent for publication

Not applicable.

Consent to participate

Not applicable.

Competing interests

Authors declare no competing interests.

Supplementary materials

Supplementary material associated with this article can be found, in the online version, at doi:10.1016/j.hlpt.2022.100602.

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